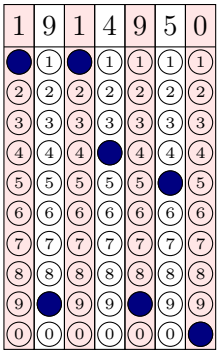
TEST VARIANTI RAQAMI: **1914950**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. $(3x-8)\sqrt{-4x^2+3x+10}=0$ tenglama nechta haqiqiy ildizga ega.

A) *2* B) *0* C) *1* D) *3*

2. (a_n) arifmetik progressiyaning dastlabki o‘n ikkita hadining yig‘indisi 432 ga teng. Agar $a_9-a_5=16$ bo‘lsa, to‘rtinchi hadini toping.

A) *22* B) *28* C) *24* D) *26*

3. Uchlari $A(10;11)$, $B(10;3)$ va $C(2;3)$ nuqtalarda bo‘lgan uchburchakning B uchidan AC tomonga BD balandlik tushirilgan. D nuqtaning koordinatalari yig‘indisini toping.

A) *9* B) $10\sqrt{2}$ C) $8\sqrt{2}$ D) *13*

4. Soddalashtiring, $(a\neq 0)$:

$$\left(\underbrace{(-2a)^3\cdot(-2a)^3\cdot\ldots\cdot(-2a)^3}_{14\ marta}\right):\left(\underbrace{(-2a)^4\cdot(-2a)^4\cdot\ldots\cdot(-2a)^4}_{10\ marta}\right)$$

A) $-8a^3$ B) $-2a$ C) $4a^2$ D) 1

5. Piyoda 3 soatda 7,8 km yo‘l yurdi. Agar piyoda shu tezlik bilan yursa, 4 soatda necha km yo‘l yuradi?

A) *10,4* B) *10,8* C) *10,5* D) *10,6*

6. Hisoblang: $\dfrac{21^3+19^3}{7^2-3^2}-21\cdot 19$

A) *3* B) *16* C) *4* D) *9*

7. $\left(2\frac{1}{4}a^2b^3\right)\cdot\left(-3\frac{1}{3}a^3b^4\right)$ ifodani soddalashtiring.

A) $-7\frac{1}{2}a^5b^7$ B) $-7\frac{1}{12}a^5b^7$ C) $-7\frac{1}{3}a^5b^7$

D) $-7\frac{1}{4}a^5b^7$

8. $2^x=2-x$ tenglama nechta haqiqiy ildizga ega?

A) *aniqlab bo‘lmaydi* B) *2*

C) *yechimga ega emas* D) *1*

9. $f(x)=24tgx-24x+6\pi+11$ funksiyaning $\left[-\frac{\pi}{4};\frac{\pi}{4}\right]$ kesmadagi eng katta qiymatini toping.

A) *35* B) *24* C) $12\sqrt{3}+13$

D) $14\sqrt{3}+11$

10. Ifodani soddalashtirting:

$$\frac{\sin 4\alpha}{1+\cos 4\alpha}\cdot\frac{\cos 2\alpha}{1+\cos 2\alpha}-\frac{\sin 2\alpha}{1+\cos 2\alpha}+1.$$

A) $\cos\alpha+1$ B) $\operatorname{tg}\alpha+1$ C) 1

D) $\sin\alpha+1$

11. $f(x)=\ln x^{x-1}$ funksiyaning hosilasini toping.

A) $f'(x)=\ln x-\dfrac{2}{x}+1$

B) $f'(x)=\ln x+1$

C) $f'(x)=\ln x-\dfrac{1}{x}+1$

D) $f'(x)=\ln x+\dfrac{1}{x}+1$

12. $y=3x^2-12x+15$ kvadrat funksiyaning qiymatlari to‘plamini aniqlang.

A) $[3;+\infty)$ B) $[7;+\infty)$ C) $[2;+\infty)$

D) $[-1;+\infty)$

13. Hisoblang: $\dfrac{\sin(\log_3 8)-\sin(\log_3 2)}{\sin(\log_3 2)+\sin(\log_3 8)}-\dfrac{\operatorname{tg}(\log_3 2)}{\operatorname{tg}(\log_3 4)}$

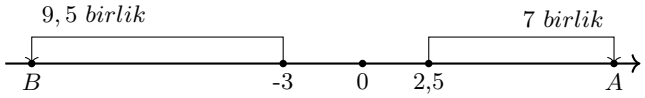
A) 2 B) -1 C) 0 D) 1

14. Aylanada 6 ta har xil nuqta belgilangan. Uchlari bu nuqtalarda bo‘lgan jami nechta har xil uchburchak chizish mumkin?

A) *20* B) *10* C) *18* D) *15*

© Davlat test markazi, 2019

15. Rasmda A va B nuqtalar son o‘qida tasvirlangan. $2A + B$ ning son qiymatini toping.



A) *9,5* B) *8* C) *12,5* D) *6,5*

16. $y = \log_3^3 x - 12$ funksiyaning qiymatlar to‘plamini toping.

A) $(12; \infty)$ B) $(-\infty; 12)$ C) $(-\infty; \infty)$ D) $(0; \infty)$

17. Tengsizlikni yeching:
$x \cdot (x-1)^2 \geq 12 \cdot (x-1)$

A) $(-\infty;\ -3] \cup [4;\ +\infty)$

B) $[-3;\ 4]$

C) $[-3;\ 1] \cup [4;\ +\infty)$

D) $(-\infty;\ -3] \cup [1;\ 4]$

18. Agar rombning bir diagonalini 50% ga kamaytirib, ikkinchi diagonalini 2 marta uzaytirilsa, rombning yuzi qanday o‘zgaradi?

A) *50% ga ortadi* B) *aniqlab bo‘lmaydi* C) *50% ga kamayadi* D) *o‘zgarmaydi*

19. Agar prizmaning qirralari soni bilan yoqlari soni yig‘indisi 38 bo‘lsa, uning uchlari sonini toping.

A) *9* B) *20* C) *18* D) *10*

20. Agar $m = 0,09$, $n = 0,16$ va $p = 0,12$ bo‘lsa,
$\left(\sqrt{\dfrac{mnp+4}{m} + 4\sqrt{\dfrac{np}{m}}}\right) : \left(2 + \sqrt{mnp}\right)$
ifodaning qiymatini toping.

A) $0,12$ B) $0,48$ C) $1\dfrac{1}{4}$ D) $3\dfrac{1}{3}$

21. $5 \cdot 19^5$ ni 6 ga bo‘lgandagi qoldiqni toping.

A) *5* B) *4* C) *2* D) *3*

22. $f(x) = (x-3)^2 + 5$ parabola uchining koordinatalari yig‘indisini toping.

A) -8 B) *2* C) *8* D) -2

23. Ishchi birinchi kuni ish normasining $\dfrac{1}{4}$ qismini bajardi. Ikkinchi kuni birinchi kunda bajarilgan ishning $\dfrac{1}{8}$ qismicha ko‘p ish bajardi. Ishchi shu ikki kunda ish normasining qancha qismini bajargan?

A) $\dfrac{5}{8}$ B) $\dfrac{17}{32}$ C) $\dfrac{19}{32}$ D) $\dfrac{9}{16}$

24. $\int\limits_0^1 x \cdot \cos x^2 dx$ integralni hisoblang.

A) $\dfrac{\cos 1}{2}$ B) 0 C) $\dfrac{\sin 1}{2}$ D) $-\dfrac{\sin 1}{2}$

25. Agar $\dfrac{a^2 + bc - ac - ab}{ab - bc + ac - c^2} + \dfrac{b}{c+b} = 2$ tenglikda $a = 18$, $b = -2$ bo‘lsa, c ning qiymatini toping.

A) *9* B) *11* C) -19 D) *20*

26. $\left(x^2 - 12x + 10\right)^2 = (3x+10)^2$ tenglamaning barcha yechimlari yig‘indisini toping.

A) *20* B) *19* C) *3* D) *24*

27. Agar qo‘shni burchaklar o‘zaro 13:17 nisbatda bo‘lsa, shu qo‘shni burchaklardan kattasini toping.

A) *68°* B) *112°* C) *78°* D) *102°*

28. Qirralari 4 dm, 5 dm va 8 dm bo‘lgan to‘g‘ri burchakli parallelepipedni tashqi tomonidan qirrasi 1 dm bo‘lgan kublarning nechtasini terib, qirralari 6 dm, 7 dm va 10 dm bo‘lgan to‘g‘ri burchakli parallelepiped yasash mumkin?

A) *192* B) *260* C) *252* D) *184*

29. $ABCD$ trapetsiyaning asoslari BC=18 va AD=50. AC dioganal o‘tkazilgan. Agar $\angle BAC = \angle ADC$ bo‘lsa, AC dioganalning uzunligini toping.

A) *30* B) *32* C) *28* D) *34*

30. [1; 200] sonlar to‘plamida nechta natural son 6 ga (qoldiqsiz) bo‘linib, 9 ga (qoldiqsiz) bo‘linmaydi?

A) *33* B) *22* C) *44* D) *11*

Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 64-bet 50	7 – 45-bet 3	13 – 40-bet 57	19 – 107-bet 4	25 – 49-bet 44
2 – 32-bet 25	8 – 55-bet 33	14 – 123-bet 10	20 – 26-bet 44	26 – 61-bet 20
3 – 119-bet 43	9 – 84-bet 65	15 – 12-bet 24	21 – 9-bet 24	27 – 90-bet 2
4 – 25-bet 25	10 – 35-bet 9	16 – 58-bet 30	22 – 72-bet 31	28 – 111-bet 35
5 – 17-bet 8	11 – 78-bet 18	17 – 68-bet 21	23 – 21-bet 49	29 – 101-bet 83
6 – 14-bet 46	12 – 75-bet 53	18 – 100-bet 75	24 – 85-bet 1	30 – 120-bet 8

© Davlat test markazi, 2019